10.4 A Molecule and a Piston

WORKSHEET

The question

STEP I 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself A ball moving at speed w hits a wall that is moving towards it at speed u, and bounces elastically. What is its speed afterwards? (Think in the wall's frame.)

STAGE 1 The speed after n bounces

Covers: the speeds

Ask yourself Think about one collision in the piston's frame. What is the particle's speed relative to the piston before and after? What is it back in the ground frame? Does bouncing off the fixed end change the speed?

STAGE 2 A recurrence for the length

Covers: the distances

Pause and try Between the nth and $(n+1)$th collisions, a time t_n, how far does the piston move? How far does the particle travel? Write both in terms of d_n and d_{n+1}, and divide one by the other to eliminate t_n.

STAGE 3 Telescoping

Covers: a formula for d_n

Ask yourself Use the recurrence again and again to write d_n as d_1 times a product of fractions. What cancels?

STAGE 4 **The case $v = u$**

Covers: the case $v = u$

Ask yourself Put $v = u$ into your formula for d_n. Then use $ut_n = d_n - d_{n+1}$ and split into partial fractions.

What's really going on

An adiabatic invariant

Multiply the speed by the length:

$$v_n d_n = (v + 2nu) \cdot \frac{v + u}{v + (2n - 1)u} d_1 \approx (v + u) d_1$$

when n is large, or when u is much smaller than v. So

$$\text{speed} \times \text{length} \approx \text{constant}.$$

If the piston moves slowly compared with the molecule, the product vd hardly changes, however far the piston moves. Quantities like this, which stay nearly constant when a system is changed slowly, are called *adiabatic invariants*. In general, the adiabatic invariant of a back-and-forth motion is $\oint p \, \mathrm{d}q$, the momentum integrated round one cycle; here that is $mv \times 2d$.

From one molecule to a gas

Since $v \propto 1/d$, the kinetic energy $\frac{1}{2}mv^2$ is proportional to $1/d^2$. For a gas of many molecules moving along one direction, temperature is proportional to the average kinetic energy, so $T \propto 1/d^2$: halving the length quadruples the temperature. In three dimensions the energy is shared between three directions of motion, and the same argument (which Feynman gives in his *Lectures*) leads to the law of adiabatic compression of a monatomic gas,

$$TV^{2/3} = \text{constant}, \qquad \text{or equivalently} \qquad pV^{5/3} = \text{constant}.$$

The exponent $\frac{5}{3}$ is the famous ratio γ of specific heats for a monatomic gas. Compressing air in a bicycle pump to a fifth of its volume would, if done quickly and with no heat

escaping, raise its absolute temperature by a factor of $5^{0.4} \approx 1.9$ (air molecules have two atoms, giving $\gamma = 1.4$ instead): from 20°C to about 280°C. The pump does not get that hot only because the compression is slow and heat leaks away.

Fermi's cosmic rays

In 1949 Enrico Fermi proposed that cosmic rays, particles arriving from space with enormous energies, are accelerated exactly as the molecule in this question is: by bouncing repeatedly off moving magnetic fields in interstellar gas clouds, gaining a little energy each time. "Fermi acceleration" is now thought to operate at the shock waves of supernova remnants, where particles cross the shock back and forth like a molecule between a piston and a wall. Adiabatic invariants also played a role in the birth of quantum mechanics: Paul Ehrenfest argued in 1916 that the quantities that are quantised must be adiabatic invariants, since a slow change cannot make a system jump between quantum states.

Try this next

1. **The invariant exactly.** Show that $v_n d_n - (v+u)d_1 = \frac{u(v+u)d_1}{v+(2n-1)u}$, and deduce that $v_n d_n \to (v+u)d_1$ as $n \to \infty$.

2. **Pulling the piston out.** Repeat the question with the piston withdrawn at speed $u < v$. Show that the particle slows down, and find $v_n d_n$. (Expanding gases cool.)

3. **Energy.** In the case $v = u$, the molecule's speed grows without limit as the piston approaches the end. Find the average force on the piston during the nth cycle (change of momentum divided by t_n) and show that it grows like n^3. What does this say about the work needed to push the piston all the way in?

4. **A bouncing ball.** A ball bounces elastically on a floor under gravity, and the floor is raised slowly. Guess an adiabatic invariant for this motion. (*Hint:* the analogue of vd is the average speed times the height of the bounce, or equivalently the energy times the time between bounces.)

Further reading

- **Richard Feynman, Robert Leighton & Matthew Sands, *The Feynman Lectures on Physics*, Vol. I** (free online), Chapter 39, "The kinetic theory of gases". The same piston argument, extended to three dimensions.
- **Stephen Blundell & Katherine Blundell, *Concepts in Thermal Physics*** (Oxford University Press). Adiabatic processes and kinetic theory at first-year level.
- **Lev Landau & Evgeny Lifshitz, *Mechanics*** (Butterworth-Heinemann), §49. A concise treatment of adiabatic invariants, for later.

The full guided solution is in Section 10.4:
https://beyondstep.keithmcnulty.org/ch10/piston.html